# Iris Flower Classification Project Report

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**Course : CSA**

**(Fundamentals in AI and ML)**

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## Introduction

This project focuses on building a Machine Learning model to classify Iris flowers into three categories:

- Setosa
- Versicolor
- Virginica

The Iris dataset is one of the most classical and simplest machine learning datasets. It contains four features related to flower measurements.

## Problem Statement

To develop a classification model that can correctly identify the species of an Iris flower based on its sepal and petal measurements.

## Functional Requirements

- Load and preprocess dataset
- Train a Machine Learning classification model
- Predict output for test data
- Evaluate accuracy
- Ability to predict species for a new given sample

## Non-functional Requirements

- High accuracy (above 90%)
- Fast execution time
- Model interpretability
- Ease of use

## System Architecture

Flow:
Dataset → Preprocessing → Train/Test Split → Model Training → Prediction → Evaluation

## Design Diagrams

✔ **Use Case Diagram**

Actors: User (Student/Developer)
Use Cases: Load data, Train model, Test model, Predict species

✔ **Workflow Diagram**

- Import libraries
- Load dataset
- Create DataFrame
- Train-test split
- Train Logistic Regression
- Evaluate accuracy
- Predict new sample

✔ **Sequence Diagram**

User → Program → ML Model → Output Prediction

✔ **Class/Component Diagram**

- Dataset component
- Model component
- Evaluation component

## Design Decisions & Rationale

**Logistic Regression chosen because**

- Simple and interpretable
- Works well on small, linearly separable datasets
- High accuracy on Iris dataset
- test_size=0.2 ensures 80-20 train–test ratio.
- max_iter=200 avoids convergence issues.

## Implementation Details

- **Libraries Used**

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import
```

- **Load Dataset**

```
iris = load_iris()
X = iris.data
y = iris.target
```

- **Create DataFrame**

```
df = pd.DataFrame(X, columns=iris.feature_names)
df['target'] = y
```

- **Train-Test Split**

```
X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.2, random_state=42)
```

- **Model Training**

```
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
```

- **Evaluation**

y_pred = model.predict(X_test)
acc = accuracy_score(y_test, y_pred)

- **Prediction of New Sample**

sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(sample)

## Python Code And Ouput:

[4]:
```
# Iris Flower Classification Project (AIML)


# 1. Import Libraries
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# 2. Load Dataset
iris = load_iris()
X = iris.data                    # features
y = iris.target                  # labels

# 3. Convert into DataFrame for easy viewing
df = pd.DataFrame(X, columns=iris.feature_names)
df['target'] = y
print("Dataset Preview:")
print(df.head())

# 4. Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# 5. Model Training
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)

# 6. Prediction
y_pred = model.predict(X_test)
```

```python
# 7. Accuracy

acc = accuracy_score(y_test, y_pred)
print("\nModel Accuracy:", acc)

# 8. Predict New Sample
sample = [[5.1, 3.5, 1.4, 0.2]]    # Example sepal/petal values
prediction = model.predict(sample)

print("\nPredicted Flower Type:", iris.target_names[prediction][0])
```

```
Dataset Preview:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   target
0       0
1       0
2       0
3       0
4       0

Model Accuracy: 1.0

Predicted Flower Type: setosa
```

## Testing Approach

- Accuracy testing using accuracy_score
- Manual prediction testing with new sample
- Verification through confusion matrix (optional)

## Challenges Faced

- Understanding Logistic Regression parameters
- Proper train-test splitting
- Model convergence issues without setting max_iter

## Key Takeaways

- Learnings & Basics of supervised learning
- Understanding logistic regression
- Dataset preprocessing
- Model evaluation techniques
- Importance of feature selection

## Future Enhancements

- Add GUI using Tkinter/Streamlit
- Train multiple ML models and compare accuracy
- Hyperparameter tuning
- Deploy model using Flask/fastAPI

## References


- Scikit-Learn Documentation
- Python Pandas Documentation
- Iris Dataset (UCI Machine Learning Repository)